

HOW TO INCORPORATE WIND, WAVES AND OCEAN CURRENTS IN THE MARINE CRAFT EQUATIONS OF MOTION

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Abstract: This paper demystifies how ocean currents together with wind and wave loads influence the marine craft equations of motion. In the literature there exists great confusion of the use of absolute and relative velocity terms when modeling rigid-body and hydrodynamic forces. The article is useful for engineers who want to simulate and predict the motions of marine craft exposed to wind, wave and ocean currents as well as control engineers evaluating the performance of marine craft control systems. The results are also very useful for testing and tuning of integral action time constants for compensation of ocean current and 2nd-order wave-induced drift forces.

Keywords: Marine systems, wind, waves, ocean currents, equations of motion

INTRODUCTION

The equations of motion for underwater vehicles, ships, ocean structures and high-speed craft are usually derived using Newtonian and Lagrangian mechanics (Fossen, 1994, 2011). The resulting models are nonlinear mass-damper-springs which include rigid-body, hydrostatic and hydrodynamic generalized forces. The motions are coupled in six degrees of freedom (DOF). Marine craft are exposed to environmental forces due to wind, waves and ocean currents, which act like forcing on the mass-damper-spring system. In hydrodynamics it is common to assume linear superposition such that forcing due to wind and waves can be treated as generalized forces that can be directly added to the nonlinear equations of motion. However, generalized forces due to ocean currents do not follow the law of linear superposition and there exists much diversity and misunderstandings in the existing literature on how to include the effects of ocean currents in the equations of motion.

This paper addresses the effect of ocean currents on marine craft in a tutorial perspective by using the concept of relative velocity, that is the velocity of the craft with respect to the ocean current, to effectively describe current-induced forces. Different properties and representations of the marine craft equations of motion are discussed and it is shown how the current velocity enters the equations. Nonlinear models for time-domain simulations and control systems design are presented using the compact vectorial notation of Fossen (1994, 2011).

1. MARINE CRAFT EQUATIONS OF MOTION

In Fossen (1991) it was shown that the coupled 6-DOF equations of a marine craft could be expressed as:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu} \quad (1)$$

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (2)$$

where $\boldsymbol{\eta} = [N, E, D, \phi, \theta, \psi]^T$ and $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ are the generalized position and velocity vectors,

respectively. The rotation and angular velocity transformation matrices between body coordinates (BODY) and the North-East-Down (NED) geographical reference frames are denoted $\mathbf{R}_b^n$ and $\mathbf{T}_\Theta$, respectively. The other quantities follow the notation of (Fossen, 1994, 2011):

$\mathbf{J}(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{R}_b^n & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{T}_\Theta \end{bmatrix}$	Euler angle velocity transformation matrix
$\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$	System inertia matrix (including added mass)
$\mathbf{C}(\boldsymbol{\nu}) = \mathbf{C}_{RB}(\boldsymbol{\nu}) + \mathbf{C}_A(\boldsymbol{\nu})$	Coriolis-centripetal matrix (including added mass)
$\mathbf{D}(\boldsymbol{\nu})$	Damping matrix
$\mathbf{g}(\boldsymbol{\eta})$	Vector of gravitational/buoyancy forces
$\boldsymbol{\tau}$	Vector of control inputs

The subscripts RB and A are used for the rigid-body and added mass terms, respectively. The rigid-body system inertia matrix $\mathbf{M}_{RB}$ satisfies:

$$\mathbf{M}_{RB} = \mathbf{M}_{RB}^\top > 0, \quad \dot{\mathbf{M}}_{RB} = \mathbf{0}_{6 \times 6}$$

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -m\mathbf{S}(\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{r}_g^b) & \mathbf{I}_b \end{bmatrix}$$

where

$$\mathbf{S}(\boldsymbol{\lambda}) = -\mathbf{S}^\top(\boldsymbol{\lambda}) = \begin{bmatrix} 0 & -\lambda_3 & \lambda_2 \\ \lambda_3 & 0 & -\lambda_1 \\ -\lambda_2 & \lambda_1 & 0 \end{bmatrix}, \quad \boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} \quad (3)$$

is the cross-product operator defined such that $\boldsymbol{\lambda} \times \mathbf{a} := \mathbf{S}(\boldsymbol{\lambda})\mathbf{a}$.

The matrix $\mathbf{C}_{RB}$ in (2) represents the Coriolis vector term $\boldsymbol{\omega}_{b/n}^b \times \mathbf{v}_{b/n}^b$ and the centripetal vector term $\boldsymbol{\omega}_{b/n}^b \times (\boldsymbol{\omega}_{b/n}^b \times \mathbf{r}_g^b)$. Contrary to the representation of $\mathbf{M}_{RB}$, it is possible to find a large number of representations for the matrix $\mathbf{C}_{RB}$. We will present some useful representations below.

Theorem 1. (Coriolis-Centripetal Matrix from $\mathbf{M}$). Consider the 6×6 constant system inertia matrix:

$$\mathbf{M} = \mathbf{M}^\top = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} > 0 \quad (4)$$

where $\mathbf{M}_{21} = \mathbf{M}_{12}^\top$. Then the Coriolis-centripetal matrix can always be parameterized such that $\mathbf{C}(\boldsymbol{\nu}) = -\mathbf{C}^\top(\boldsymbol{\nu})$ by choosing:

$$\mathbf{C}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} \\ -\mathbf{S}(\mathbf{M}_{11}\boldsymbol{\nu}_1 + \mathbf{M}_{12}\boldsymbol{\nu}_2) \\ -\mathbf{S}(\mathbf{M}_{21}\boldsymbol{\nu}_1 + \mathbf{M}_{22}\boldsymbol{\nu}_2) \end{bmatrix} \quad (5)$$

with $\boldsymbol{\nu}_1 := [u, v, w]^\top$ and $\boldsymbol{\nu}_2 := [p, q, r]^\top$.

Proof: Sagatun and Fossen (1991).

The rigid-body Coriolis and centripetal matrix $\mathbf{C}_{RB}(\boldsymbol{\nu})$ can always be parametrized such that it is skew-symmetrical:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^\top(\boldsymbol{\nu}), \quad \forall \boldsymbol{\nu} \in \mathbb{R}^6 \quad (6)$$

The skew-symmetric property is very useful when designing nonlinear motion control systems since the quadratic form $\boldsymbol{\nu}^\top \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} \equiv 0$. This is exploited in energy-based designs where Lyapunov functions play a key role. There exists several parametrizations (Fossen and Fjellstad, 1995) that satisfy (6).

Lagrangian parametrization: Application of Theorem 1 with $\mathbf{M} = \mathbf{M}_{RB}$ yields the following expression:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_g^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_g^b) - \mathbf{S}(\mathbf{I}_b\boldsymbol{\nu}_2) \end{bmatrix} \quad (7)$$

Linear velocity-independent parametrization: By using the cross-product property $\mathbf{S}(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2 = -\mathbf{S}(\boldsymbol{\nu}_2)\boldsymbol{\nu}_1$, it is possible to move $\mathbf{S}(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2$ from $\mathbf{C}_{RB}^{\{12\}}$ to $\mathbf{C}_{RB}^{\{11\}}$ in (7). This gives an expression for $\mathbf{C}_{RB}(\boldsymbol{\nu})$ that is independent of linear velocity $\boldsymbol{\nu}_1$:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{r}_g^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b\boldsymbol{\nu}_2) \end{bmatrix} \quad (8)$$

Remark 1. Formula (8) is useful when irrotational ocean currents enter the equations of motion since $\mathbf{C}_{RB}(\boldsymbol{\nu})$ does not depend on the linear velocity $\boldsymbol{\nu}_1$ (uses only angular velocity $\boldsymbol{\nu}_2$ and lever arm $\mathbf{r}_g^b$). This is discussed in Section 3.2.

Gravitational and buoyancy forces for surface vessels show a linear behavior $\mathbf{g}(\boldsymbol{\eta}) = \mathbf{G}\boldsymbol{\eta}$ where

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -Z_z & 0 & -Z_\theta & 0 \\ 0 & 0 & 0 & -K_\phi & 0 & 0 \\ 0 & 0 & -M_z & 0 & -M_\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (9)$$

For underwater vehicles

$$\mathbf{g}(\boldsymbol{\eta}) = \begin{bmatrix} (W - B) \sin(\theta) \\ -(W - B) \cos(\theta) \sin(\phi) \\ -(W - B) \cos(\theta) \cos(\phi) \\ -(y_g W - y_b B) \cos(\theta) \cos(\phi) \\ (z_g W - z_b B) \sin(\theta) \\ -(x_g W - x_b B) \cos(\theta) \sin(\phi) \\ + (z_g W - z_b B) \cos(\theta) \sin(\phi) \\ + (x_g W - x_b B) \cos(\theta) \cos(\phi) \\ - (y_g W - y_b B) \sin(\theta) \end{bmatrix} \quad (10)$$

where the gravitational and buoyancy forces act

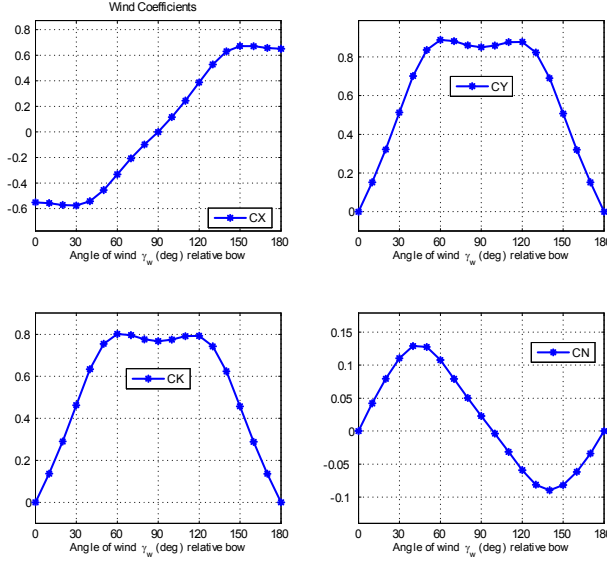


Fig. 1. Wind coefficients C_X, C_Y, C_K and C_N for a research vessel.

through the centers of gravity (CG) and buoyancy (CB) defined by the vectors $\mathbf{r}_g^b := [x_g, y_g, z_g]^T$ and $\mathbf{r}_b^b := [x_b, y_b, z_b]^T$, respectively.

2. SUPERPOSITION OF WIND AND WAVE-INDUCED FORCES

For control systems design it is common to assume the *principle of superposition* when considering wind and wave-induced forces such that (2) takes the following form:

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau} \quad (11)$$

where $\boldsymbol{\tau}_{\text{wind}} \in \mathbb{R}^6$ and $\boldsymbol{\tau}_{\text{wave}} \in \mathbb{R}^6$ represent the generalized forces due to wind and waves.

2.1 Wind forces and moments

Wind can be defined as the movement of air relative to the surface of the Earth. For a marine craft moving at a forward speed, the wind forces and moments:

$$\boldsymbol{\tau}_{\text{wind}} = \frac{1}{2}\rho_a V_{rw}^2 \begin{bmatrix} C_X(\gamma_{rw})A_{Fw} \\ C_Y(\gamma_{rw})A_{Lw} \\ C_Z(\gamma_{rw})A_{Fw} \\ C_K(\gamma_{rw})A_{Lw}H_{Lw} \\ C_M(\gamma_{rw})A_{Fw}H_{Fw} \\ C_N(\gamma_{rw})A_{Lw}L_{oa} \end{bmatrix} \quad (12)$$

are functions of relative wind speed V_{rw} and angle of attack γ_{rw} according to:

$$V_{rw} = \sqrt{u_{rw}^2 + v_{rw}^2} \quad (13)$$

$$\gamma_{rw} = -\text{atan2}(v_{rw}, u_{rw}) \quad (14)$$

which both are functions of the relative velocities:

$$u_{rw} = u - u_w \quad (15)$$

$$v_{rw} = v - v_w \quad (16)$$

The nondimensional wind coefficients C_X, C_Y, C_Z, C_K, C_M and C_N are usually computed using $h = 10$ m as reference height while ρ_a is the air density. The frontal and lateral projected wind areas are denoted by A_{Fw} and A_{Lw} while L_{oa} is the length over all. The mean heights of the areas A_{Fw} and A_{Lw} are denoted by H_{Lw} and H_{Fw} , respectively. Figure 1 shows four wind coefficients for a typical research vessel (Blendermann, 1994). Wind coefficients for other vessels are found in Fossen (2011) and references therein.

2.2 Wave-induced forces and moments

The first- and second-order wave forces for varying wave directions β_i and wave frequencies ω_k are denoted $\tilde{\tau}_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i)$ and $\tilde{\tau}_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i)$ where $\text{dof} \in \{1, 2, 3, 4, 5, 6\}$. The normalized force *response amplitude operators* (RAOs) are complex variables given by (WAMIT Inc., 2010):

$$F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i) = \left| \frac{\tilde{\tau}_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i)}{\rho g A_k} \right| e^{j\angle \tilde{\tau}_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i)}$$

$$F_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i) = \left| \frac{\tilde{\tau}_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i)}{\rho g A_k^2} \right| e^{j\angle \tilde{\tau}_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i)}$$

The output from the hydrodynamic code is usually an ASCII file containing RAOs in table format. Let us denote the imaginary and real parts of the force RAOs by; $\text{Im}_{\text{wave1}}\{\text{dof}\}(k, i)$ and $\text{Re}_{\text{wave1}}\{\text{dof}\}(k, i)$. The amplitudes and phases for different frequencies ω_k and wave directions β_i for the first-order wave-induced forces can be computed according to the formulae:

$$\left| F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i) \right| = \left(\text{Im}_{\text{wave1}}\{\text{dof}\}(k, i)^2 + \text{Re}_{\text{wave1}}\{\text{dof}\}(k, i)^2 \right)^{1/2} \quad (17)$$

$$\angle F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i) = \text{atan2}(\text{Im}_{\text{wave1}}\{\text{dof}\}(k, i), \text{Re}_{\text{wave1}}\{\text{dof}\}(k, i)) \quad (18)$$

The amplitudes and phases for the second-order mean forces are:

$$\left| F_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i) \right| = \text{Re}_{\text{wave2}}\{\text{dof}\}(k, i) \quad (19)$$

$$\angle F_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i) = 0 \quad (20)$$

Since the first- and second-order wave forces are represented in terms of the complex variables $F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta_i)$ and $F_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta_i)$, the responses for sinusoidal excitations can be computed using different wave spectra. A frequently used family of wave spectra is:

$$S(\omega) = A\omega^{-5} \exp(-B\omega^{-4}) \quad (21)$$

where different values for A and B are used. These values depend on geographical location and wind speed (see Chapter 8.2, Fossen 2011).

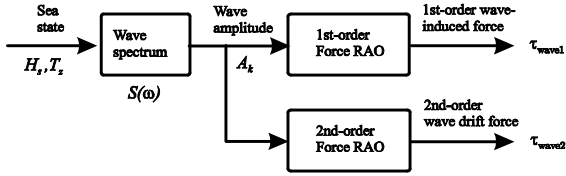


Fig. 2. Computation of first- and second-order wave-induced forces from force RAOs.

When computing the wave-induced forces, linear superposition is employed as illustrated in Fig. 2. The relationship between the a wave spectrum $S(\omega_k)$ and the wave amplitude A_k for a wave component k is (Faltinsen, 1990):

$$\frac{1}{2}A_k^2 = S(\omega_k)\Delta\omega \quad (22)$$

where $\Delta\omega$ is a constant difference between the frequencies. Let the wave-induced forces in 6 DOF be denoted by the vectors:

$$\begin{aligned} \tau_{\text{wave1}} &= [\tau_{\text{wave1}}^{\{1\}}, \tau_{\text{wave1}}^{\{2\}}, \dots, \tau_{\text{wave1}}^{\{6\}}]^T \\ \tau_{\text{wave2}} &= [\tau_{\text{wave2}}^{\{1\}}, \tau_{\text{wave2}}^{\{2\}}, \dots, \tau_{\text{wave2}}^{\{6\}}]^T \end{aligned}$$

For the no spreading case, the wave direction $\beta =$ constant such that:

$$\begin{aligned} \tau_{\text{wave1}}^{\{\text{dof}\}} &= \sum_{k=1}^N \rho g \left| F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta) \right| \\ &A_k \cos(\omega_e(U, \omega_k, \beta)t + \angle F_{\text{wave1}}^{\{\text{dof}\}}(\omega_k, \beta) + \epsilon_k) \end{aligned} \quad (23)$$

$$\begin{aligned} \tau_{\text{wave2}}^{\{\text{dof}\}} &= \sum_{k=1}^N \rho g \left| F_{\text{wave2}}^{\{\text{dof}\}}(\omega_k, \beta) \right| \\ &A_k^2 \cos(\omega_e(U, \omega_k, \beta)t + \epsilon_k) \end{aligned} \quad (24)$$

where

$$\omega_e(U, \omega_k, \beta) = \omega_k - \frac{\omega_k^2}{g}U \cos(\beta) \quad (25)$$

is the encounter frequency. The assumption that $\beta =$ constant can be relaxed to model spreading of the main wave propagation direction (see Chapter 8.3, Fossen 2011).

3. EQUATIONS OF MOTION INCLUDING OCEAN CURRENTS

Ocean currents are horizontal and vertical circulation systems of ocean waters produced by gravity, wind friction and water density variation in different parts of the ocean. Besides *wind-generated currents*, the heat exchange at the sea surface together with salinity changes, develop an additional sea current component, usually referred to as *thermohaline currents*.

The oceans are conveniently divided into two water spheres, the cold and warm water sphere. Since the Earth is rotating, the Coriolis force will try

to turn the major currents to the East in the northern hemisphere and West in the southern hemisphere. Finally, the major ocean circulations will also have a tidal component arising from planetary interactions and gravity. In coastal regions and fjords, tidal components can reach very high speeds, in fact speeds of 2–3 m/s or more have been measured.

The forces on a marine craft due to ocean currents can be accounted for by replacing the generalized velocity vector in the hydrodynamic terms with relative velocities:

$$\mathbf{v}_r = \mathbf{v} - \mathbf{v}_c \quad (26)$$

where $\mathbf{v}_c \in \mathbb{R}^6$ is the velocity of the ocean current expressed in BODY.

Definition 1. (Irrotational fluid). The generalized ocean current velocity of an *irrotational fluid* is:

$$\mathbf{v}_c = [\underbrace{u_c, v_c, w_c}_{\mathbf{v}_c^b}, 0, 0, 0]^T \quad (27)$$

where $\mathbf{v}_c^b = [u_c, v_c, w_c]^T$ is the linear velocity.

The ocean current linear velocity vector satisfies:

$$\mathbf{v}_c^n = \mathbf{R}_b^n(\Theta_{nb})\mathbf{v}_c^b \quad (28)$$

where $\Theta_{nb} = [\phi, \theta, \psi]^T$ are the Euler angles between BODY and NED, and $\mathbf{R}_b^n(\Theta_{nb}) \in SO(3)$ is the corresponding rotation matrix.

Definition 2. (Irrotational constant ocean current). An irrotational constant ocean current in NED satisfies:

$$\dot{\mathbf{v}}_c^n = \dot{\mathbf{R}}_b^n(\Theta_{nb})\mathbf{v}_c^b + \mathbf{R}_b^n(\Theta_{nb})\dot{\mathbf{v}}_c^b := \mathbf{0} \quad (29)$$

where

$$\dot{\mathbf{R}}_b^n(\Theta_{nb}) = \mathbf{R}_b^n(\Theta_{nb})\mathbf{S}(\omega_{b/n}^b) \quad (30)$$

This implies that the ocean current linear velocity vector in BODY coordinates is given by:

$$\dot{\mathbf{v}}_c^b = -\mathbf{S}(\omega_{b/n}^b)\mathbf{v}_c^b \quad (31)$$

3.1 Equations of Motion including Ocean Currents

In order to simulate irrotational ocean currents and their effect on marine craft motion, the following model can be applied:

$$\begin{aligned} &\underbrace{\mathbf{M}_{RB}\dot{\mathbf{v}} + \mathbf{C}_{RB}(\mathbf{v})\mathbf{v} + \mathbf{g}(\boldsymbol{\eta})}_{\text{rigid-body and hydrostatic terms}} \\ &+ \underbrace{\mathbf{M}_A\dot{\mathbf{v}}_r + \mathbf{C}_A(\mathbf{v}_r)\mathbf{v}_r + \mathbf{D}(\mathbf{v}_r)\mathbf{v}_r}_{\text{hydrodynamic terms}} \\ &= \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau} \end{aligned} \quad (32)$$

where

$$\mathbf{v}_r = \begin{bmatrix} \mathbf{v}^b - \mathbf{v}_c^b \\ \omega_{b/n}^b \end{bmatrix} \quad (33)$$

is the relative velocity vector. Notice that the

rigid-body kinetics is independent of the ocean current.

3.2 Equations of Relative Velocity

It is possible to simplify (32) by exploiting the structure of $\mathbf{C}_{RB}(\boldsymbol{\nu}_r)$.

Theorem 2. If the Coriolis and centripetal matrix $\mathbf{C}_{RB}(\boldsymbol{\nu}_r)$ is parametrized independent of linear velocity $\boldsymbol{\nu}_1 = [u, v, w]^\top$, for instance by using (8), and the ocean current is irrotational and constant (Definition 2), the rigid-body kinetics satisfies (Hegrenæs, 2010):

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} \equiv \mathbf{M}_{RB}\dot{\boldsymbol{\nu}}_r + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r \quad (34)$$

PROOF. Since the Coriolis and centripetal matrix represented by (8) is independent of linear velocity $\boldsymbol{\nu}_1 = [u, v, w]^\top$, it follows that $\mathbf{C}_{RB}(\boldsymbol{\nu}_r) = \mathbf{C}_{RB}(\boldsymbol{\nu})$. The property:

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}}_c + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_c = \mathbf{0} \quad (35)$$

is proven by expanding the matrices $\mathbf{M}_{RB}$ and $\mathbf{C}_{RB}(\boldsymbol{\nu}_r)$, and corresponding acceleration and velocity vectors according to:

$$\begin{aligned} & \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -m\mathbf{S}(\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{r}_g^b) & \mathbf{I}_b \end{bmatrix} \begin{bmatrix} -\mathbf{S}(\boldsymbol{\omega}_{b/n}^b)\mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix} \\ & + \begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{b/n}^b) & -m\mathbf{S}(\boldsymbol{\omega}_{b/n}^b)\mathbf{S}(\mathbf{r}_g^b) \\ m\mathbf{S}(\mathbf{r}_g^b)\mathbf{S}(\boldsymbol{\omega}_{b/n}^b) & -\mathbf{S}(\mathbf{I}_b\boldsymbol{\omega}_{b/n}^b) \end{bmatrix} \begin{bmatrix} \mathbf{v}_c^b \\ \mathbf{0}_{3 \times 1} \end{bmatrix} = \mathbf{0} \end{aligned}$$

Finally, it follows that:

$$\begin{aligned} \mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} &= \mathbf{M}_{RB}[\dot{\boldsymbol{\nu}}_r + \dot{\boldsymbol{\nu}}_c] \\ &+ \mathbf{C}_{RB}(\boldsymbol{\nu}_r)[\boldsymbol{\nu}_r + \boldsymbol{\nu}_c] \\ &= \mathbf{M}_{RB}\dot{\boldsymbol{\nu}}_r + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r \end{aligned} \quad (36)$$

Theorem 2 when applied to (32) gives the differential equations:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}_r + \begin{bmatrix} \mathbf{v}_c^n \\ \mathbf{0} \end{bmatrix} \quad (37)$$

$$\mathbf{M}\dot{\boldsymbol{\nu}}_r + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau} \quad (38)$$

where $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ and $\mathbf{C}(\boldsymbol{\nu}_r) = \mathbf{C}_{RB}(\boldsymbol{\nu}_r) + \mathbf{C}_A(\boldsymbol{\nu}_r)$. Notice that only $\boldsymbol{\nu}_r$ and not $\boldsymbol{\nu}$ is used in (38) if compared to (32). The model (37)–(38) includes the bias $\dot{\mathbf{v}}_c^n = \mathbf{0}$ at the *kinematic* level while (32) models drift due to ocean currents at the *kinetic* level using $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$.

3.3 Equations of Motion for Zero Speed

For low-speed applications such as DP, ocean currents and damping can be modeled by three *current coefficients* C_X, C_Y and C_N . These can be experimentally obtained using scale models in wind tunnels. The resulting forces are measured on the model, which is restrained from moving.

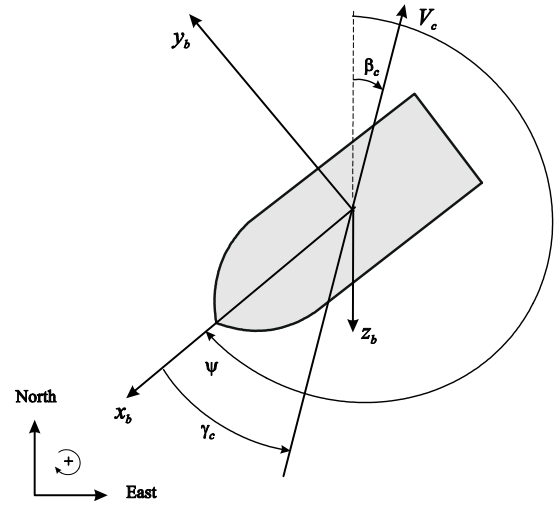


Fig. 3. Current speed V_c , current direction β_c and current angle of attack γ_c relative bow.

In many textbooks and papers, for instance Blendenmann (1994), wind and current coefficients are defined relative to the bow using a counter clockwise rotation γ_c (see Figure 3). The current forces on a marine craft at rest can be expressed in terms of the area-based current coefficients C_X, C_Y and C_N as:

$$X_{\text{current}} = \frac{1}{2}\rho A_{Fc} C_X(\gamma_c) V_c^2 \quad (39)$$

$$Y_{\text{current}} = \frac{1}{2}\rho A_{Lc} C_Y(\gamma_c) V_c^2 \quad (40)$$

$$N_{\text{current}} = \frac{1}{2}\rho A_{Lc} L_{oa} C_N(\gamma_c) V_c^2 \quad (41)$$

where V_c is the speed of the ocean current. The frontal and lateral projected currents areas are denoted A_{Fc} and A_{Lc} , respectively while L_{oa} is the length over all and ρ is the density of water.

For vehicles at rest and motions limited to surge, sway and yaw, ocean currents are linearly superimposed according to:

$$\begin{aligned} \mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) &= \\ &\boldsymbol{\tau}_{\text{current}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau} \end{aligned}$$

where $\boldsymbol{\tau}_{\text{current}} = [X_{\text{current}}, Y_{\text{current}}, N_{\text{current}}]^\top$.

The current coefficients can also be used at forward speed $U > 0$ and related to the surge resistance, cross-flow drag and the Munk moment used in maneuvering theory by using the concept of relative velocity (see Chapter 7.3, Fossen 2011).

4. OCEAN CURRENT SIMULATION MODELS

Let the ocean current speed be denoted by V_c while its direction relative to the moving craft is expressed in terms of two angles: *angle of attack* α_c and *sidelip angle* β_c as shown in Figure 4. For computer simulations the ocean current speed

and direction can be generated by using first-order *Gauss–Markov processes*:

$$\dot{V}_c + \mu_1 V_c = w_1 \quad (42)$$

$$\dot{\alpha}_c + \mu_2 \alpha_c = w_2 \quad (43)$$

$$\dot{\beta}_c + \mu_3 \beta_c = w_3 \quad (44)$$

where w_i ($i = 1, 2, 3$) are zero-mean Gaussian white noise processes and $\mu_i \geq 0$ ($i = 1, 2, 3$) are constants. If $\mu_1 = \mu_2 = \mu_3 = 0$, the models reduce to a *random walks*, corresponding to time integration of *white noise*. A saturating element is usually used in the integration process to limit the current speed to:

$$V_{\min} \leq V_c(t) \leq V_{\max} \quad (45)$$

The direction of the current can also be fixed by specifying constant values for α_c and β_c .

3-D Irrotational Ocean Current Model: A 3-D irrotational ocean current model is obtained by transforming the ocean current speed V_c and directions (α_c, β_c) from FLOW axes to NED velocities:

$$\begin{aligned} \mathbf{v}_c^n &= \mathbf{R}_{y, \alpha_c}^\top \mathbf{R}_{z, -\beta_c}^\top \begin{bmatrix} V_c \\ 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} V_c \cos(\alpha_c) \cos(\beta_c) \\ V_c \sin(\beta_c) \\ V_c \sin(\alpha_c) \cos(\beta_c) \end{bmatrix} \end{aligned} \quad (46)$$

where the principal rotations $\mathbf{R}_{y, \alpha_c}$ and $\mathbf{R}_{z, -\beta_c}$ are recognized as:

$$\mathbf{R}_{y, \alpha} = \begin{bmatrix} \cos(\alpha) & 0 & \sin(\alpha) \\ 0 & 1 & 0 \\ -\sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix} \quad (47)$$

$$\mathbf{R}_{z, -\beta} = \begin{bmatrix} \cos(\beta) & \sin(\beta) & 0 \\ -\sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (48)$$

This expression can be transformed from NED to BODY using the Euler angle rotation matrix $\mathbf{R}_b^n$. Consequently,

$$\begin{bmatrix} u_c \\ v_c \\ w_c \end{bmatrix} = \mathbf{R}_b^n (\boldsymbol{\Theta}_{nb})^\top \begin{bmatrix} V_c \cos(\alpha_c) \cos(\beta_c) \\ V_c \sin(\beta_c) \\ V_c \sin(\alpha_c) \cos(\beta_c) \end{bmatrix} \quad (49)$$

2-D Irrotational Ocean Current Model: For motions in the horizontal plane, the 3-D equations (49) reduce to:

$$u_c = V_c \cos(\beta_c - \psi) \quad (50)$$

$$v_c = V_c \sin(\beta_c - \psi) \quad (51)$$

for $\alpha_c = 0$ and $\phi = \theta = 0$. Consequently,

$$V_c = \sqrt{u_c^2 + v_c^2} \quad (52)$$

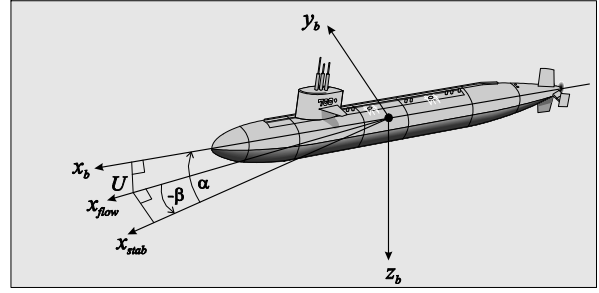


Fig. 4. Angle of attack and sideslip angle for a marine craft.

5. CONCLUSIONS

A tutorial on how to include models for wind, waves and ocean currents for marine craft has been presented. The concept of equations of relative motions is used to model the effect of ocean currents while wind and wave-induced forces are added under the assumption of linear superposition. The article is intended for control engineers who want to simulate and predict the motions of marine craft exposed to wind, wave and ocean currents and use time-series to evaluate the performance of control systems.

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